ModelIt!

Teacher's Guide

The Light You're Seeing Is Already Old

When You Look at a Star, You're Seeing the Past

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-ESS1-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-ESS1-1

Support an argument that differences in the apparent brightness of the sun compared to other stars is due to their relative distances from Earth.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use models to understand relationships between distance and brightness.
Disciplinary Core Idea	**ESS1.A The Universe and Its Stars** — The sun is a star that appears larger and brighter than other stars because it is closer.
Crosscutting Concept	**Scale, Proportion, and Quantity** — Students use appropriate scales to understand astronomical distances.

CAST Testing Alignment

MULTIPLE CHOICE:

Star A is twice as far from Earth as Star B. If both stars have the same luminosity, how will their apparent brightness compare?

CONSTRUCTED RESPONSE:

Explain why the sun appears so much brighter than other stars even though many other stars are actually much larger and more luminous.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

When we look at the night sky, we're looking at the past. Light takes time to travel across the vast distances of space, so we never see stars as they are 'now.'

Light Speed

Light travels at 186,000 miles per second - fast, but the universe is enormous. Light from the nearest star takes 4 years to reach us.

Inverse Square Law

As distance doubles, brightness decreases by a factor of 4. Triple the distance, 9 times dimmer. This is why nearby stars appear so much brighter.

The Sun's Proximity

The sun is about 93 million miles away. The next nearest star is about 25 trillion miles away. The sun appears bright because it's incomparably close.

Hubble Deep Field

In 1995, Hubble pointed at 'empty' sky for 10 days. It revealed 10,000 galaxies, each containing billions of stars. The farthest light was 12 billion years old.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify distance, luminosity, light travel time, and apparent brightness as components.

E — Establish Relationships
Students establish that distance increases travel time (positive) but decreases brightness (negative), while luminosity increases brightness (positive).

V — Visualize & Model
Students build the starlight model with competing influences on apparent brightness.

E — Evaluate Outcomes
Students compare sun, dim nearby star, and distant supergiant scenarios.

R — Revise & Extend
Students add star color/temperature to predict luminosity.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The light you're seeing is already old. Some of it is ancient."*

Watch For

Create a sense of wonder.

Slide 2: Learning Objectives

2 min

Say: *"Today we model why stars look the way they do."*

Watch For

Define light year.

Slide 3: Big Question

2 min

Say: *"How do we figure out why some stars look brighter?"*

Watch For

Show travel times to different stars.

Slide 4: LEVER Framework

2 min

Say: *"We'll model the night sky."*

Watch For

Preview activities.

Slide 5: Activity 1: Components

8 min

Say: *"Two forces affect apparent brightness - one helps, one hurts."*

Watch For

Set up the competition.

Slide 6: Activity 2: Connections

8 min

Say: *"Distance is NEGATIVE for brightness. This is the key."*

Watch For

Use streetlight analogy.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's compare the sun to distant supergiants."*

Watch For

Students run all three scenarios.

Slide 8: Discoveries

5 min

Say: *"You're looking at the past. Some stars might not exist anymore."*

Watch For

Let existential wonder land.

Slide 9: STEM Challenge

10 min

Say: *"Design a mission to observe a distant star."*

Watch For

Discuss what we can learn from light.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Star Distance, Star Actual Luminosity
- **INTERNAL:** Light Travel Time, Apparent Brightness

Relationships

- **Star Distance to Light Travel Time:** POSITIVE (+) — Farther stars mean light takes longer to reach us.
- **Star Distance to Apparent Brightness:** NEGATIVE (-) — Farther stars appear dimmer because light spreads out.
- **Star Luminosity to Apparent Brightness:** POSITIVE (+) — More luminous stars appear brighter at the same distance.

Key Discoveries

- The sun looks brightest because it's CLOSE, not because it's special
- Distance has a NEGATIVE effect on apparent brightness
- Very luminous stars can be visible from thousands of light years away
- Light travel time means we're always seeing the past

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Bigger stars are always brighter"	Apparent brightness depends on BOTH luminosity AND distance. A massive but distant star can appear dimmer than a small nearby one.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The night sky shows current universe"	All starlight is from the past. Farther = older. We literally cannot see the present state of distant objects.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"The sun is a special star"	The sun is actually a fairly average star. It appears special only because it's so much closer than any other star.